

Locally Convex Topological Vector Spaces

Definitions and Examples

Definition: A topological vector space (TVS) is a vector space X with a topology $\mathcal{T}$ such that, with respect to this topology, the maps

- (i) $X \times X \rightarrow X, (x, y) \mapsto x + y$
- (ii) $\mathbb{F} \times X \rightarrow X, (\alpha, x) \mapsto \alpha x$

are continuous with respect to $\mathcal{T}$.

Example: If $\mathcal{P}$ is a family of seminorms on X , and $\mathcal{T}$ is the topology on X with subbasic sets $U(x_0, p, \varepsilon) = \{x \in X : p(x - x_0) < \varepsilon\}$, then $(X, \mathcal{T})$ is a topological vector space.

Definition: A locally convex topological vector space (LCS) is a TVS equipped with a family of seminorms $\mathcal{P}$ that satisfy

$$\bigcap_{p \in \mathcal{P}} \{x : p(x) = 0\} = \{0\}.$$

That is, the family $\mathcal{P}$ separates the points of X .

Proposition: Let X be a TVS, and let p be a seminorm on X . The following are equivalent:

- (a) p is continuous;
- (b) $\{x \in X : p(x) < 1\}$ is open;
- (c) $0 \in \{x \in X : p(x) < 1\}^\circ$;
- (d) $0 \in \{x \in X : p(x) \leq 1\}^\circ$;
- (e) p is continuous at 0;
- (f) there is a continuous seminorm q on X such that $p \leq q$.

Proof. We see immediately that (a) implies (b) since the set is equal to the inverse image of the open sub-set $[0, 1) \subseteq [0, \infty)$. Similarly, if $p^{-1}([0, 1))$ is open, then it is equal to its own interior, and since $0 \in p^{-1}([0, 1))$, it follows that there is a subbasic open neighborhood of 0 contained in $p^{-1}([0, 1))$, giving (c). Similarly, we obtain (d) by the fact that $0 \in p^{-1}([0, 1))^\circ \subseteq p^{-1}([0, 1])^\circ$.

Next, we see that (d) implies that for every $\varepsilon > 0$, $0 \in p^{-1}([0, \varepsilon])^\circ$, so if $(x_i)_i$ is a net in X converging to 0 and $\varepsilon > 0$, there is some i_0 such that for all $i \geq i_0$, $x_i \in \{x : p(x) \leq \varepsilon\}$. That is, $p(x_i) \leq \varepsilon$ for all $i \geq i_0$, whence p is continuous at 0. This gives (e).

If p is continuous at 0, then if $x_i \rightarrow x$ is a convergent net in X , the reverse triangle inequality gives $|p(x) - p(x_i)| \leq p(x_i - x)$, whence $p(x_i) \rightarrow p(x)$, so p is continuous. This gives that (e) implies (a).

Taking $q = p$ yields (a) implies (f). We will show that (f) implies (e). If $x_i \rightarrow 0$ in X , then $q(x_i) \rightarrow 0$, but since $0 \leq p(x_i) \leq q(x_i)$, it follows that $p(x_i) \rightarrow 0$. $\square$

Example:

- (a) Let X be a completely regular topological space, and let $C(X)$ be the set of all continuous functions from X to $\mathbb{F}$. If K is compact in X , we may define a seminorm $p_K(f) = \sup_{x \in K} |f(x)|$. Then,

$$\{p_K : K \subseteq X \text{ compact}\}$$

forms a separating family of seminorms for $C(X)$.

- (b) Let G be an open subset of $\mathbb{C}$. If $H(G)$ is the set of holomorphic functions on G , then the same seminorms $\{p_K : K \subseteq G \text{ compact}\}$ forms a locally compact space whose topology is uniform convergence on compact subsets.

- (c) Let X be a normed vector space. For each $\varphi \in X^*$, let $p_\varphi(x) = |\varphi(x)|$. Then, $\mathcal{P} = \{p_\varphi : \varphi \in X^*\}$ forms a separating family of seminorms for X . The topology these seminorms induce is called the *weak topology* on X . It is often denoted $\sigma(X, X^*)$.
- (d) If X is a normed space, then for each $x \in X$, we may define a seminorm $p_x : X^* \rightarrow [0, \infty)$ by taking $p_x(\varphi) = |\varphi(x)|$. This defines a separating family of seminorms $\mathcal{P} = \{p_x : x \in X\}$, and the topology on X^* induced by these seminorms is called the *weak**, or w^* , topology. It is often denoted $\sigma(X^*, X)$.

Next, we focus on an examination of the “convexity” part of the notion of locally convex topological vector spaces.

Definition: A subset $A \subseteq X$ is said to be convex if for all $v, w \in A$, the segment

$$[v, w] := \{tv + (1-t)w : 0 \leq t \leq 1\}$$

is contained in A .

Note that the intersection of any family of convex subsets is convex.

Definition: If $A \subseteq X$ is any subset, then the *convex hull* of A , denoted $\text{conv}(A)$, is the intersection of all the convex subsets that contain A . If X is a TVS, the *closed convex hull* of A , denoted $\overline{\text{conv}}(A)$, is the intersection of all the closed convex sets that contain A .

Given any linear map $T : X \rightarrow Y$ of real vector spaces, and $C \subseteq X$ a convex subset of Y , then $T^{-1}(C)$ is convex in X . If X^* is the dual of any complex vector space, then since X is also a real vector space, it then follows that the sets

$$\begin{aligned} A_0 &= \{x : |f(x)| \leq 1\} \\ A_1 &= \{x : \text{Re}(f(x)) \leq 1\} \\ A_3 &= \{x : \text{Re}(f(x)) > 1\} \end{aligned}$$

are all convex.

Proposition: Let X be a TVS, and let A be a convex subset of X . Then,

- (a) $\overline{A}$ is convex;
 (b) if $a \in A^\circ$ and $b \in \overline{A}$, then

$$[a, b] := \{tb + (1-t)a : 0 \leq t < 1\}$$

is contained in A° .

Proof. Let $a \in A$, $b \in \overline{A}$, and let $0 \leq t \leq 1$. Let $(x_i)_i$ be a net in A such that $x_i \rightarrow b$. Then, $tx_i + (1-t)a \rightarrow tb + (1-t)a$. In particular, this means that if $b \in \overline{A}$ and $a \in A$, then $[a, b] \subseteq \overline{A}$. So, given any $a, b \in \overline{A}$, we have that the net $(x_i)_i \rightarrow b$ is such that $[x_i, a] \subseteq A$, so it follows that $[b, a] \subseteq \overline{A}$, whence $\overline{A}$ is convex.

For (b), fix $0 < t < 1$, and set $c = tb + (1-t)a$ for some $a \in A^\circ$ and $b \in \overline{A}$. Since $a \in A^\circ$, there is an open neighborhood V of 0 such that $a + V \subseteq A$. Therefore, for any $d \in A$, we have

$$\begin{aligned} A &\supseteq td + (1-t)(a + V) \\ &= t(d - b) + tb + (1-t)(a + V) \\ &= (t(d - b) + (1-t)V) + c. \end{aligned}$$

We hope to find some $d \in A$ such that $0 \in t(d - b) + (1-t)V$, which we call U . The preceding inclusion then shows that $c \in A^\circ$, since U is an open neighborhood of 0 with $c + U \subseteq A$.

Finding such a d is equivalent to finding d such that $0 \in \frac{1}{t}(1-t)V + (d - b)$, or $d \in b - \frac{1}{t}(1-t)V$. Yet, we know that $0 \in -\frac{1}{t}(1-t)V$ by the assumption that V is an open neighborhood of 0. Since $b \in \overline{A}$, it follows that $b - \frac{1}{t}(1-t)V \cap A \neq \emptyset$ by the definition of closure, whence such a $d \in A$ exists. $\square$

Corollary: If $A \subseteq X$, then $\overline{\text{conv}}(A) = \overline{\text{conv}(A)}$.

Definition: A set $A \subseteq X$ is called balanced if for any $|\alpha| \leq 1$ and any $x \in A$, we have $\alpha x \in A$.

We say a set A is *absorbing* if for any $x \in X$, there is some $\varepsilon > 0$ such that $tx \in A$ for all $0 \leq t < \varepsilon$. We say A is absorbing at a if for any $x \in X$, there is $\varepsilon > 0$ such that $a + tx \in A$ for all $0 \leq t < \varepsilon$.

Given a seminorm p on V , the set $V = \{x : p(x) < 1\}$ is a convex balanced set that is absorbing at each of its points. We will show that the converse actually holds — that is, a vector space is a LCS if and only if the set of all open, convex, balanced subsets of X forms a neighborhood basis at 0.

Proposition: Suppose X is a vector space over F , and V is a nonempty convex balanced set that is absorbing at each of its points. Then, there is a unique seminorm p on X such that

$$V = \{x \in X : p(x) < 1\}.$$

Proof. Let

$$p(x) = \inf\{t \geq 0 : x \in tV\}.$$

Since V is absorbing, it follows that $X = \bigcup_{n=1}^{\infty} nV$, so this is a well-defined quantity. Furthermore, $p(0) = 0$.

To show that $p(\alpha x) = |\alpha|p(x)$, we suppose $\alpha \neq 0$, and since V is balanced,

$$\begin{aligned} p(\alpha x) &= \inf\{t \geq 0 : \alpha x \in tV\} \\ &= \inf\left\{t \geq 0 : x \in \frac{t}{\alpha}V\right\} \\ &= \inf\left\{t \geq 0 : x \in \frac{t}{|\alpha|}V\right\} \\ &= |\alpha| \inf\left\{\frac{t}{|\alpha|} : x \in \frac{t}{|\alpha|}V\right\} \\ &= |\alpha|p(x). \end{aligned}$$

We now observe that if $\alpha, \beta \geq 0$, then for any $a, b \in V$, we have

$$\begin{aligned} \alpha a + \beta b &= (\alpha + \beta) \left(\frac{\alpha}{\alpha + \beta} a + \frac{\beta}{\alpha + \beta} b \right) \\ &\in (\alpha + \beta)V, \end{aligned}$$

since V is convex. For any $x, y \in X$, take $p(x) = \alpha$ and $p(y) = \beta$, and let $\delta > 0$. Then, $x \in (\alpha + \delta)V$ and $y \in (\beta + \delta)V$ by the definition of the infimum, so that $x + y \in (\alpha + \beta + 2\delta)V$. Therefore, we have $p(x + y) \leq \alpha + \beta + 2\delta$, which holds for all $\delta > 0$, so that $p(x + y) \leq p(x) + p(y)$.

Finally, we will have to show that $V = \{x : p(x) < 1\}$. If $p(x) = \alpha < 1$, then for any $\alpha < \beta < 1$, we have $x \in \beta V \subseteq V$, which emerges from the fact that V is balanced. Thus, $V \supseteq \{x : p(x) < 1\}$. If $x \in V$, then $p(x) \leq 1$, and since V is absorbing at x , there is some $\varepsilon > 0$ such that for $0 < t < \varepsilon$, we have $y = x + tx \in V$. Yet, this means $x = \frac{1}{1+t}y$ with $y \in V$, so that $p(x) = \frac{1}{1+t}p(y) \leq \frac{1}{1+t} < 1$. $\square$

The seminorm p is called the Minkowski function for V , or the *gauge* of V .

Proposition: Let X be a TVS, and let $\mathcal{U}$ be the collection of open, convex, balanced subsets of X . Then, X is locally convex if and only if $\mathcal{U}$ is a basis for the neighborhood system at 0.

Proof. Suppose X is locally convex with topology defined by the family of seminorms $\mathcal{P}$. Then, we observe that each of the subbasic open neighborhoods of 0,

$$\{U(0, p, \varepsilon) : \varepsilon > 0, p \in \mathcal{P}\},$$

are open, convex, and balanced; since the set of all finite intersections of all these subbasic open neighborhoods yields a neighborhood basis for 0, and finite intersections of open, balanced, convex subsets are open, balanced, and convex, so it follows that $\mathcal{U}$ contains the set of all finite intersections of all these subbasic open

neighborhoods, whence $\mathcal{U}$ contains a basis for the neighborhood system at 0. In particular, this means $\mathcal{U}$ also must generate the neighborhood system at 0, so $\mathcal{U}$ forms a basis for the neighborhood system at 0.

Now, suppose $\mathcal{U}$ forms a basis for the neighborhood system at 0. Observe that an open, convex, balanced subset of V contains 0, and is itself absorbing. We claim that the family of Minkowski functionals

$$\mathcal{P} = \{p_V(\cdot - z) : z \in X, V \in \mathcal{U}\}$$

generates the topology for X . □

Weak Topologies and Extremal Structure

C^* -Algebras

Spectral Theory

Spectral theory is primarily concerned with representations of abelian C^* -algebras. Consider a normal operator N on H . Then, $C^*(N)$ is an abelian C^* -algebra, and the continuous functional calculus $f \mapsto f(N)$ yields a $*$ -isomorphism of $C(\sigma(N))$ onto $C^*(N)$. In particular, $f \mapsto f(N)$ is a representation of the abelian C^* -algebra $C^*(N)$.

Definition: Let X be a set, Σ a σ -algebra on X , and H a Hilbert space. A *spectral measure* for (X, Σ, H) is a map $E: \Sigma \rightarrow B(H)$ satisfying

- (a) for each $\Delta \in \Sigma$, $E(\Delta)$ is a projection;
- (b) $E(\emptyset) = 0$ and $E(X) = 1$;
- (c) $E(\Delta_1 \cap \Delta_2) = E(\Delta_1)E(\Delta_2)$ for any $\Delta_1, \Delta_2 \in \Sigma$;
- (d) given pairwise disjoint $\{\Delta_n\}_{n=1}^\infty$, we have

$$E\left(\bigcup_{n=1}^\infty \Delta_n\right) = \sum_{n=1}^\infty E(\Delta_n),$$

where the latter is an SOT-convergent sum of pairwise orthogonal projections converging to the projection onto the supremum $\bigwedge_{n=1}^\infty E(\Delta_n)H$.

Recall that the SOT and WOT are defined by the seminorms

- $p_h(A) = \|Ah\|$ for the SOT, and
- $p_{h,k}(H) = |\langle Ah, k \rangle|$ for the WOT.

Example: If (X, Ω, μ) is a σ -finite measure space, and $\phi \in L^\infty(X, \mu)$, then $M_\phi \in B(L^2(X, \mu))$ can be defined by $M_\phi(f) = \phi f$.

A net $(\phi_i)_{i \in I}$ in $L^\infty(X, \mu)$ is w^* -convergent to ϕ if and only if $M_{\phi_i} \rightarrow M_\phi$ in WOT. This follows from the fact that if $f, g \in L^2(X, \mu)$, then $f\bar{g} \in L^1(X, \mu)$, where we have $L^1(X, \mu)^* = L^\infty(X, \mu)$, so that if $(\phi_i)_i \rightarrow \phi$ weak*, then

$$\begin{aligned} \langle M_{\phi_i} f, g \rangle &= \int \phi_i(f\bar{g}) d\mu \\ &\rightarrow \int \phi f\bar{g} d\mu \\ &= \langle M_\phi f, g \rangle, \end{aligned}$$

while the converse emerges from the fact that any element $f \in L^1(X, \mu)$ can be written as $f = g_1 \bar{g}_2$.

Example: If E is a spectral measure for (X, Ω, H) , then $E^{(n)}(\Delta) = E(\Delta)^{(n)}$ (under the diagonal embedding) is a spectral measure for $(X, \Omega, H^{(n)})$.

We will now use spectral measures to define a representation of a C^* -algebra.

Lemma: Let E be a spectral measure on (X, Ω, H) , and let $g, h \in H$. Then,

$$E_{g,h}(\Delta) = \langle E(\Delta)g, h \rangle$$

defines a countably additive measure on Ω with total variation bounded above by $\|g\|\|h\|$.

Proof. For countable additivity, we note the classic result that a sequence of projections $(P_n)_n \rightarrow P$ converges in WOT if and only if it converges in SOT. In particular, this means that if $\{\Delta_n\}_{n \in \mathbb{N}}$ is a sequence of disjoint subsets of X , then we have a sequence of projections

$$\left(\sum_{k=0}^n E(\Delta_k) \right)_n \rightarrow \sum_{k=0}^{\infty} E(\Delta_k)$$

converging in SOT, hence converging in WOT. Yet, this means that

$$\begin{aligned} \left\langle \left(\sum_{k=0}^n E(\Delta_k) \right) g, h \right\rangle &= \sum_{k=0}^n \langle E(\Delta_k)g, h \rangle \\ &\rightarrow \left\langle \left(\sum_{k=0}^{\infty} E(\Delta_k) \right) g, h \right\rangle, \end{aligned}$$

so that

$$\sum_{k=0}^{\infty} \langle E(\Delta_k)g, h \rangle = \left\langle E \left(\bigcup_{k=0}^{\infty} \Delta_k \right) g, h \right\rangle,$$

yielding our desired result.

If we have a finite set $\Delta_1, \dots, \Delta_n$ of pairwise disjoint sets, we choose $|\alpha_j| = 1$ such that $\left| \langle E(\Delta_j)g, h \rangle \right| = \alpha_j \langle E(\Delta_j)g, h \rangle$, whence

$$\begin{aligned} \sum_{j=1}^n |\mu(\Delta_j)| &= \left\langle \sum_{j=1}^n \alpha_j E(\Delta_j)g, h \right\rangle \\ &\leq \left\| \sum_{j=1}^n E(\Delta_j) \alpha_j g \right\| \|h\|. \end{aligned}$$

Notice that the family of $E(\Delta_j)$ are pairwise orthogonal projections, meaning that by Pythagoras's Theorem and the additivity of the spectral measure, we have

$$\begin{aligned} \left\| \sum_j E(\Delta_j) \alpha_j g \right\|^2 &= \sum_{j=1}^n \|E(\Delta_j)g\|^2 \\ &= \left\| E \left(\bigcup_{j=1}^n \Delta_j \right) g \right\|^2 \\ &\leq \|g\|^2, \end{aligned}$$

whence $\sum_{j=1}^n |\mu(\Delta_j)| \leq \|g\|\|h\|$. Since this holds for any finite collection of subsets, we have $\|\mu\|_{TV} \leq \|g\|\|h\|$. $\square$

We next concern ourselves with establishing “integration” with respect to the spectral measure.

Proposition: Let E be a spectral measure for (X, Ω, H) , and let $\phi: X \rightarrow \mathbb{C}$ be a bounded Ω -measurable function. Then, there is a unique operator $A \in B(H)$ such that for any $\varepsilon > 0$ and partition $\{\Delta_1, \dots, \Delta_n\}$ with $\sup_{x, x' \in \Delta_k} |\phi(x) - \phi(x')| < \varepsilon$ for each k , we have

$$\left\| A - \sum_{k=1}^n \phi(x_k) E(\Delta_k) \right\| < \varepsilon$$

for any $x_k \in \Delta_k$.

Proof. For each $g, h \in H$, define a (bounded) sesquilinear form

$$B(g, h) = \int \phi dE_{g, h},$$

with $|B(g, h)| \leq \|\phi\|_{L^\infty} \|g\| \|h\|$. Then, by the structure of bounded sesquilinear forms, there is a unique operator A such that $B(g, h) = \langle Ag, h \rangle$ for all $g, h \in H$.

Consider a partition $\{\Delta_1, \dots, \Delta_n\}$ satisfying the conditions in the proposition. For any arbitrary vectors $g, h \in H$ and $x_k \in \Delta_k$, we have

$$\begin{aligned} \left| \langle Ag, h \rangle - \sum_{k=1}^n \phi(x_k) \langle E(\Delta_k)g, h \rangle \right| &= \left| \sum_{k=1}^n \int_{\Delta_k} (\phi(x) - \phi(x_k)) dE_{g, h} \right| \\ &\leq \sum_{k=1}^n \int_{\Delta_k} |\phi(x) - \phi(x_k)| d|E_{g, h}| \\ &\leq \varepsilon \int d|E_{g, h}| \\ &\leq \varepsilon \|g\| \|h\|. \end{aligned}$$

This yields the desired result. □

For the operator A in the above statement, we write

$$A = \int \phi dE.$$

That is, for any $g, h \in H$ and bounded Ω -measurable function ϕ on X , we have

$$\left\langle \left(\int \phi dE \right) g, h \right\rangle = \int \phi dE_{g, h}.$$

References

- [Ped89] Gert K. Pedersen. *Analysis Now*. Vol. 118. Graduate Texts in Mathematics. Springer-Verlag, New York, 1989, pp. xiv+277. ISBN: 0-387-96788-5. DOI: [10.1007/978-1-4612-1007-8](https://doi.org/10.1007/978-1-4612-1007-8). URL: <https://doi.org/10.1007/978-1-4612-1007-8>.
- [Con90] John B. Conway. *A Course in Functional Analysis*. Second. Vol. 96. Graduate Texts in Mathematics. Springer-Verlag, New York, 1990, pp. xvi+399. ISBN: 0-387-97245-5.